Artificial Intelligence (2026 Spring)

UVA CS_4710-002

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Course Description

How do self-driving cars navigate on the road, or chatbots (ChatGPT, Gemini, Claude, ...) determine what to say? At the core of the technological transformation is the increasing ability of machines to learn, reason, and interact intelligently with complex data, humans, and the world. Artificial Intelligence (AI) is the core technology behind this transformation, allowing everything from our mobile phones and smart homes to robots and medical diagnostic tools to understand complex information and act intelligently.

But AI is about so much more. Have you ever wondered: Can machines truly learn, understand, and even create? What separates today's AI from mere automation? And how do we ensure these intelligent technologies serve humanity ethically? This course will explore and apply the core concepts of AI, from intelligent agents and search to machine learning and generative AI. You'll gain hands-on experience through class exercises and homework assignments. You'll solidify your understanding of AI concepts with weekly review questions and midterms. A significant component of this course is the project, where you'll have the opportunity to design, implement, and evaluate an AI system for a problem you found at UVA/Charlottesville.

Prerequisite: CS 2150 or CS 3100 with a grade of C- or better.

Time and Location

- TuTh 11:00 am - 12:15 pm
- Olsson 009

Teaching Team

- **Instructor: Yen-Ling Kuo**
  - Email: ylkuo@virginia.edu
  - Office Hour: Wednesdays, 11-11:45 am or by appointment, Olsson 261
- **TA: Xuhui (Joshua) Kang**
  - Email: qhv6ku@virginia.edu
  - Office Hour: Fridays, 5-6pm, Rice 232 or Zoom (see Canvas)
- **TA: Chen Gong**
  - Email: fzv6en@virginia.edu
  - Office Hour: Thursdays, 4-5pm, Rice 442
- **TA: Rui Shen**
  - Email: kqn9mt@virginia.edu
  - Office Hour: Tuesdays, 4-5pm, Rice 232

Course Format

The course will be delivered through lectures. In addition to two lectures a week, students will have weekly review questions, 5 programming homework assignments, two midterm exams, and a final project to practice the concepts presented in the lectures.

Textbook (optional)

- Russell and Norvig. Artificial Intelligence: A Modern Approach. 3rd Edition. 2010. (or 4th Edition)

Details and grading policy

- Weekly review questions: 5%
  - Weekly review questions will help you review the key concepts of lectures and course material. Review questions will be offered at the end of most weeks in a multiple-choice format on Gradescope.
  - You can review them at your own pace after class. Questions will be available for a week, no extension. Only your **top five** scores will be used in the final grade.
- Participation: 5%
  - As AI is evolving very quickly, we encourage you to have continuous participation **within and outside of lectures** in any format to keep engaging in or learning about AI. This includes, but is not limited to, the following activities.
    - In-class activities (responses submitted through online forms), AI news/tips sharing (up to two students each lecture), Piazza discussions (including questions and answering about AI, not logistics), scribe lectures (up to two scribers per lecture), and participate in AI activities (hackathons, talks/seminars, etc.) outside of class.
  - You can participate in any format that works best for you. At least ten activities are needed to have full participation credit in the final grade.
- Homework: 30%
  - Homework assignments are designed to provide you with individual hands-on practice to translate theoretical knowledge into working code (in Python). You will be implementing various types of AI agents to play a game like Minesweeper! In each assignment, we provide you with a description of agent design and skeleton code. You will apply the foundational concepts and critical thinking you learned to complete the AI agents.
  - Late submission is allowed for three days after the deadline with a **20% deduction per day**. We will automatically grant **two slip days** for each student throughout the semester.
    - Slip day: Using slip days avoids the late penalty. You can use your two slip days in one homework (two days) or two homeworks (one day each homework).

- Midterm: 30%
  - The midterm exams will serve as a comprehensive assessment to check your understanding of core AI concepts and methods. It will be an open-note exam, including multiple-choice questions and short-answer questions. Questions will focus on the concept, algorithm, and simple calculations.
- Course project: 30%
  - The course project challenges you to integrate various AI concepts into a cohesive solution for a real-world problem, a problem you found in UVA/Charlottesville! Working in teams, you will navigate the full problem-solving lifecycle, from identifying appropriate AI methods and leveraging libraries to researching relevant resources. This hands-on experience is critical for developing the confidence and practical skills needed to tackle complex AI challenges beyond the classroom.

Final project guidelines

- Topic: Solving problems in UVA or Charlottesville with AI
  - Identify a problem you or students face in UVA/Charlottesville, design an AI agent, and apply the AI or generative AI methods taught in the class to implement the designed AI agent to solve the problem.
- Details:

  🔬 **Final Project Guideline**

Communication

- Canvas: https://canvas.its.virginia.edu/courses/167352
  - Course materials (slides), Course recordings, Project submissions
- Piazza: https://piazza.com/class/mka0bmcy26z1qb
  - Announcements, Q&A, Discussions
- Gradescope: https://www.gradescope.com/courses/1220922
  - Weekly review questions, Homework submissions

(Tentative) Syllabus

Date	Topics	Homework & Project	Notes
Week 1 Tue, Jan 13	Introduction, Overview of AI		AIMA Ch1
Week 1 Thu, Jan 15	Intelligent Agents	HW1 out: rule-based agent	AIMA Ch2
	Topic 1: Problem Solving		
Week 2 Tue, Jan 20	Problem solving by search: Uninformed search		AIMA 3.1-3.4
Week 2 Thu, Jan 22	Problem solving by search: Informed search		AIMA 3.5, 3.6
Week 3 Tue, Jan 27	Logic & Inference	HW2 out: search-based agent	AIMA Ch7
	Topic 2: Reasoning with Uncertainty		
Week 3 Thu, Jan 29	**Rivanna Access, Agentic Framework Tutorial**		
Week 4 Tue, Feb 3	Quantifying uncertainty		AIMA Ch13
Week 4 Thu, Feb 5	Probabilistic reasoning: Bayes Nets, Inference by Enumeration	Project: Team and topic registration	AIMA 14.1-14.4
Week 5 Tue, Feb 10	Probabilistic reasoning over time: Markov Process, Inference in Temporal Models	HW3 out - Bayes Net agent	AIMA 15.1-15.3
	Topic 3: Reinforcement Learning		
Week 5 Thu, Feb 12	Reinforcement learning: Markov Decision Process		AIMA 17.1
Week 6 Tue, Feb 17	Reinforcement learning: Policy, Value, Value Iteration		AIMA 17.2
Week 6 Thu, Feb 19	Reinforcement learning: Model-based, Model-free, Q-learning	Project: Agent storyboard due	AIMA Ch21

Week 7 Tue, Feb 24	*Midterm Review*	HW4 out: RL agent	
Week 7 Thu, Feb 26	**Midterm 1**		
Week 8 Tue, Mar 3	*Spring recess*		
Week 8 Thu, Mar 5	*Spring recess*		
	Topic 4: Learning from examples		
Week 9 Tue, Mar 10	Learning from examples: Naive Bayes, Train/Validation/Test		AIMA 20.1-20.2
Week 9 Thu, Mar 12	Learning from examples: Features, Perceptron	Project: Technical proposal due	AIMA 18.6.3, 18.7.1-18.7.2
Week 10 Tue, Mar 17	Learning from examples: Logistic Regression, Regularization		AIMA 18.4, 18.6.4
Week 10 Thu, Mar 19	Learning from examples: Neural Networks	HW5 out: Neural Network	AIMA 18.7
Week 11 Tue, Mar 24	Learning from examples: Non-parametric models, kNN		AIMA 18.8.1
Week 11 Thu, Mar 26	**Rivanna GPU, Neural Network Tutorial**		
	Topic 5: AI Applications & Future		
Week 12 Tue, Mar 31	Application: Robotics		
Week 12 Thu, Apr 2	Project working time + milestone feedback	Project: milestone due on Apr 1 for feedback on this date	
Week 13 Tue, Apr 7	**Midterm 2**		

Week 13 Thu, Apr 9	Project working time + milestone feedback	Project: milestone due on Apr 8 for feedback on this date	
Week 14 Tue, Apr 14	Application/GenAI: Natural language processing, Language generation		
Week 14 Thu, Apr 16	Application/GenAI: Computer vision, Image generation		
Week 15 Tue, Apr 21	GenAI: Generative agents, Issues		
Week 15 Thu, Apr 23	Responsible AI: Explainability, Bias, Fairness	Project: Video due on 4/25 (Sat)	
Week 16 Tue, Apr 28	*Course end* Project Spotlight		
Week 16 Thu, Apr 30		Project: Final report due	

Generative AI Policy

In this course, we will explore different topics in AI, and of course, how to use generative AI tools is of interest. We also have some opportunities for you to practice with these tools. I encourage you to explore generative AI tools to help your learning, but very importantly, use them **with a critical and responsible view**. That means

- DO NOT simply copy the review questions, homework assignments, and lecture videos to ask AI to generate answers and code to submit without working on them yourself.
- DO NOT use AI to directly generate reports without thinking about the project and working on it yourself.
- DO discuss with AI to clarify or explain the concepts you do not understand well, and if uncertain, ask and discuss on Piazza or during office hours!

- DO brainstorm ideas with AI, but critically think about whether the ideas make sense or how you may use or implement them.
- DO use AI to improve your writing, such as proofreading, grammar/spellchecking, suggesting organization of content, etc, but review them to learn what is edited and if you agree with them.

The goal of using AI is to help you learn and provide you with more ways to learn at your own pace, but the goal is not to outsource your brain to AI and not to do your coursework (this is exactly replacing yourself with an AI!)

If you use any generative AI tools in your homework assignments and projects in any way, you are required to acknowledge and cite all generative AI use/contributions.

- Create an appendix at the end of your submission with the following information:
  - The name of the platforms you used
  - The exact prompts or conversations you had with AI
  - The raw AI output (if it answers)
  - Any iterations or adaptations you have to improve the outputs
  - Explain how you use the output or what you find the AI fell short on

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Statements

Honor

I trust every student in this course to fully comply with all of the provisions of the University's Honor Code. By enrolling in this course, you have agreed to abide by and uphold the Honor System of the University of Virginia, as well as the following policies specific to this course.

- All graded assignments must be pledged.
- All suspected violations will be forwarded to the Honor Committee, and you may, at my discretion, receive an immediate zero on that assignment regardless of any action taken by the Honor Committee.

Please let me know if you have any questions regarding the course Honor policy. If you believe you may have committed an Honor Offense, you may wish to file a Conscientious Retraction by calling the Honor Offices at (434) 924-7602. For your retraction to be considered valid, it must, among other things, be filed with the Honor Committee before you are aware that the act in question has come under suspicion by anyone. More information can be found at http://honor.virginia.edu. Your Honor representatives can be found at: http://honor.virginia.edu/representatives.

Students with disabilities or learning needs

It is my goal to create a learning experience that is as accessible as possible. If you anticipate any issues related to the format, materials, or requirements of this course, please meet with me outside of class so we can explore potential options. Students with disabilities may also wish to work with the Student Disability Access Center (SDAC) to discuss a range of options to removing barriers in this course, including official accommodations. We are fortunate to have an SDAC advisor, Courtney MacMasters, physically located in Engineering. You may email her at cmacmasters@virginia.edu to schedule an appointment. For general questions please visit the SDAC website: *sdac.studenthealth.virginia.edu*. If you have already been approved for accommodations through SDAC, please send me your accommodation letter and meet with me so we can develop an implementation plan together.

Religious accommodations

It is the University's long-standing policy and practice to reasonably accommodate students so that they do not experience an adverse academic consequence when sincerely held religious beliefs or observances conflict with academic requirements.

Students who wish to request academic accommodation for a religious observance should submit their request to me by email as far in advance as possible. Students who have questions or concerns about academic accommodations for religious observance or religious beliefs may contact the University's Office for Equal Opportunity and Civil Rights (EOCR) at *UVAEOCR@virginia.edu* or 434-924-3200.

Harassment, Discrimination, and Interpersonal Violence

The University of Virginia is dedicated to providing a safe and equitable learning environment for all students. If you or someone you know has been affected by power-based personal violence, more information can be found on the UVA Sexual Violence website that describes reporting options and resources available - *www.virginia.edu/sexualviolence*.

The same resources and options for individuals who experience sexual misconduct are available for discrimination, harassment, and retaliation. UVA prohibits discrimination and harassment based on age, color, disability, family medical or genetic information, gender identity or expression, marital status, military status, national or ethnic origin, political affiliation, pregnancy (including childbirth and related conditions), race, religion, sex, sexual orientation, or veteran status. UVA policy also prohibits retaliation for reporting such behavior.

If you witness or are aware of someone who has experienced prohibited conduct, you are encouraged to submit a report to Just Report It (justreportit.virginia.edu) or contact EOCR, the office of Equal Opportunity and Civil Rights.

If you would prefer to disclose such conduct to a confidential resource where what you share is not reported to the University, you can turn to Counseling & Psychological Services ("CAPS") and Women's Center Counseling Staff and Confidential Advocates (for students of all genders).

As your professor and as a person, know that I care about you and your well-being and stand ready to provide support and resources as I can. As a faculty member, I am a responsible employee, which means that I am required by University policy and by federal law to report certain kinds of conduct that you report to me to the University's Title IX Coordinator. The Title IX Coordinator's job is to ensure that the reporting student receives the resources and support that they need, while also determining whether further action is necessary to ensure survivor safety and the safety of the University community.

Support for your career development

Engaging in your career development is an important part of your student experience. For example, presenting at a research conference, attending an interview for a job or internship, or participating in an extern/shadowing experience are not only necessary steps on your path but are also invaluable lessons in and of themselves. I wish to encourage and support you in activities related to your career development. To that end, please notify me by email as far in advance as possible to arrange for appropriate accommodations.

Student support team

You have many resources available to you when you experience academic or personal stresses. In addition to your professor, the School of Engineering and Applied Science has staff members located in Thornton Hall who you can contact to help manage academic or personal challenges. Please do not wait until the end of the semester to ask for help!

Learning

Lisa Lampe, Assistant Dean for Undergraduate Affairs

Georgina Nembhard, Director of Student Success

Courtney MacMasters, Accessibility Specialist

Free tutoring is available for most classes.

Health and Wellbeing

Kelly Garrett, Assistant Dean of Students, Student Safety and Support